Task 14 - VPC networking setup

Objective

Design and implement a **multi-VPC AWS network environment** across **two regions** and establish connectivity using **VPC Peering**.

Problem Statement

ABC Technologies is expanding its AWS infrastructure across multiple locations and wants to build a simple inter-office network design in the cloud.

The company has one **HQ office** and four branch offices: **A, B, C, and D**. The requirement is to host the **HQ office in** `us-east-1` and all four branch offices in `us-east-2`.

Instead of using a single VPC with subnetting, the company wants to create **separate VPCs for each office** and connect them using **VPC Peering**. This will help the team understand cross-region connectivity and multi-VPC design in AWS.

The **HQ office** should use a **Class A private IP-based network**, while the branch offices should use **Class B private IP-based networks**. The HQ office must contain **2 subnets**:

- One subnet for **servers**
- One subnet for **management**

Each branch office should be planned based on the required host count.

As a DevOps intern, your task is to design the network, create the VPCs in AWS, connect them using peering, deploy EC2 instances for testing, and verify connectivity.

Network Requirements

Region 1: HQ Office

- **Location:** `us-east-1`
- **Network Type:** Class A private IP
- **Subnets Required:** 2
  - Server subnet
  - Management subnet
- **Required Hosts:**
  - Server subnet: **200 hosts**

- Management subnet: **200 hosts**

Region 2: Branch Offices

- **Location:** `us-east-2`
- **Network Type:** Class B private IP
- **Branch Offices:**
  - Office A → **50 hosts**
  - Office B → **50 hosts**
  - Office C → **50 hosts**
  - Office D → **50 hosts**

Task Requirements

1] Network Design

Create a network design for:

- 1 VPC for **HQ** in `us-east-1`
- 4 separate VPCs for **Office A, B, C, and D** in `us-east-2`

Ensure:

- HQ uses **Class A private IP addressing**
- Branch offices use **Class B private IP addressing**
- No IP overlap between VPCs
- Host requirements are met for each office

Do **not** use subnetting for branch offices beyond what is required for the VPC design.
Students should decide the final CIDR ranges themselves.

2] AWS VPC Environment Setup

Create the required VPCs in AWS:

- HQ VPC in `us-east-1`
- Office A VPC in `us-east-2`
- Office B VPC in `us-east-2`
- Office C VPC in `us-east-2`
- Office D VPC in `us-east-2`

For HQ:

- Create 2 subnets:
  - Server subnet
  - Management subnet

For branch offices:

- Create the required VPC and subnet structure based on the host requirement

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3] VPC Peering

Set up **cross-region VPC peering** between:

- HQ ↔ Office A
- HQ ↔ Office B
- HQ ↔ Office C
- HQ ↔ Office D

Update route tables appropriately so communication works between HQ and each office.

4] EC2 Deployment and Connectivity Testing

- Launch EC2 instances in:
  - HQ VPC
    - HQ Server subnet
    - HQ Management subnet
  - Each office VPC
- Configure security groups to allow testing
- Verify connectivity between HQ and branch office instances using basic commands such as:

```
telnet <private-ip> 22
```

or equivalent connectivity checks.

5] Documentation

Document:

- Chosen network ranges
- Region-wise VPC setup
- Peering configuration

- Route table updates
- EC2 placement
- Connectivity test results

Expected Deliverables

Submit the following:

- Network design diagram showing HQ and office VPCs
- Screenshot of VPC peering and route table configuration
- EC2 deployment and connectivity proof
- Requirement analysis (Confluence link)
- Technical documentation (Confluence link)
- Prepare a short document explaining the difference between the OSI model and the TCP/IP model, and describe 10 networking devices along with their uses. (Confluence link)
  - Also include a short video (2–3 mins) explaining the same documentation.

Time to Complete

3 Hours